

The 5k — O-F-O Triplet Injector Design

ENGINE PARAMETERS (from CEA / main.py)

Thrust (SL / vac)	2.50 / 3.07 kN
Chamber pressure P_c	20.0 bar
Mixture ratio O/F	2.00
Propellants	LOX / RP-1
mdot total	1.0308 kg/s
mdot fuel (RP-1)	0.3436 kg/s
mdot ox (LOX)	0.6872 kg/s
Chamber ID at injector	96.77 mm (R = 48.40 mm)

INJECTOR DESIGN (O-F-O TRIPLET, N = 20 ELEMENTS)

Element type	O-F-O triplet
Element count	20
Total orifices	60 (20 fuel + 40 ox)
dP_inj (both sides)	4.0 bar (20% of P_c)
Discharge coefficient	Cd = 0.65 (K = 2.37)
Per-element mdot	51.5 g/s

ORIFICE SPECIFICATIONS

	Fuel (central)	Ox (outer x2)
Diameter	1.154 mm	1.056 mm each
Length (L/D = 6)	6.92 mm	6.34 mm
Velocity	31.6 m/s	26.5 m/s
Reynolds	17,200	168,000
mdot per orifice	17.2 g/s	17.2 g/s

IMPINGEMENT GEOMETRY

Total angle 2-theta	60 deg
Per-jet angle theta	30 deg from element axis
Free length L_imp	6.34 mm from face
Ox-orifice radial offset	+/- 3.66 mm from fuel centerline
Element width	7.32 mm

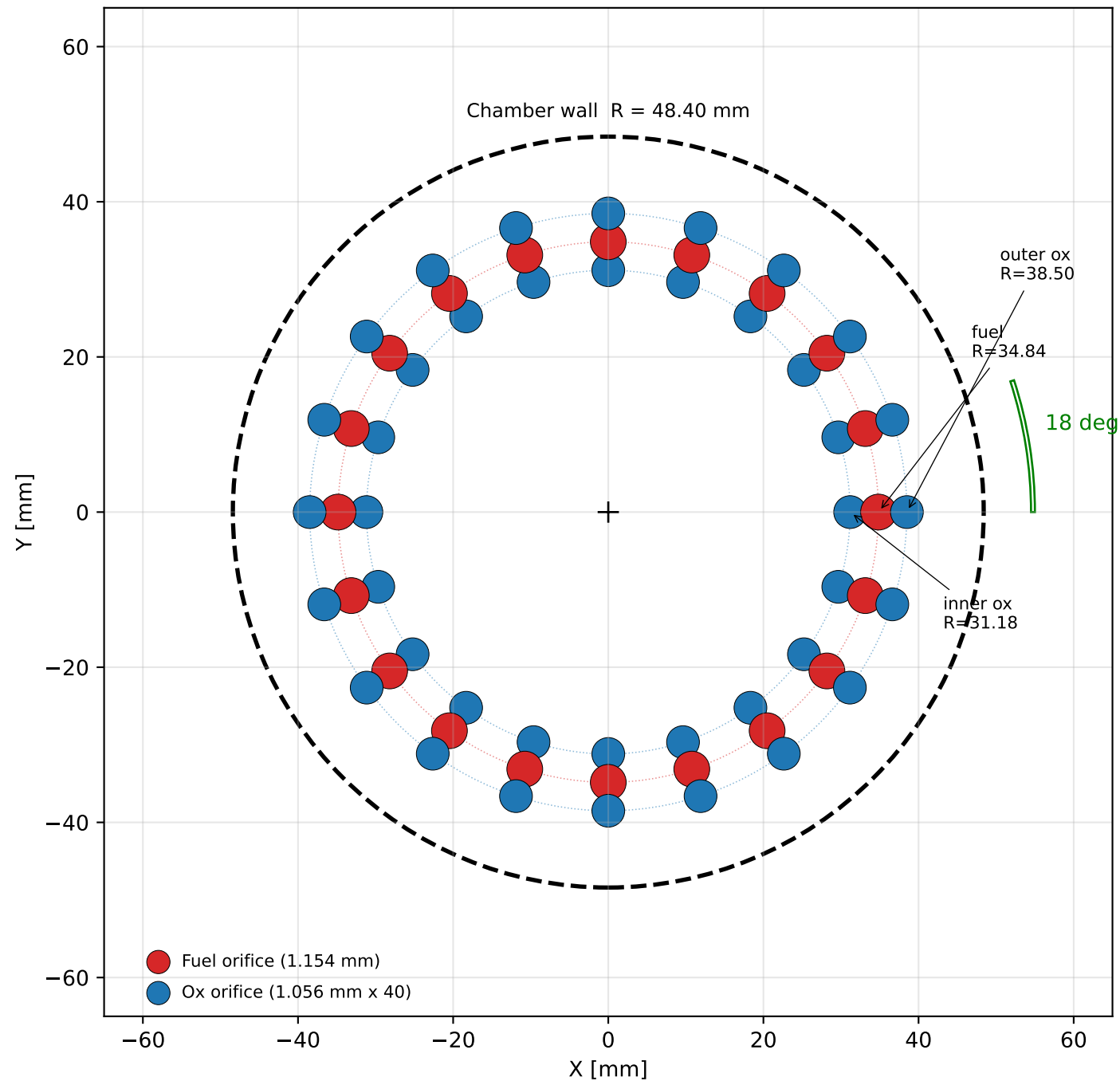
FACE COOLING -- LOX (20 radial channels) *** CORRECTED ***

Coolant	LOX (NOT fuel: RP-1 is regen-heated to ~500 K & coking-limited)
Cross-section	1.5 x 1.5 mm (SLM depowder OK)
Total channels	20 (between element pairs)
Total flow area	45 mm^2
Velocity	13.4 m/s
Reynolds	117,000 (turbulent)
h_cool	36.2 kW/(m^2K)
dT_film	110 K
Max face temp	253 K (huge margin vs 800 K)
LOX bulk T rise	90 -> 115 K (22 K subcooled)
dP across face channels	0.35 bar

FLUID PATHS

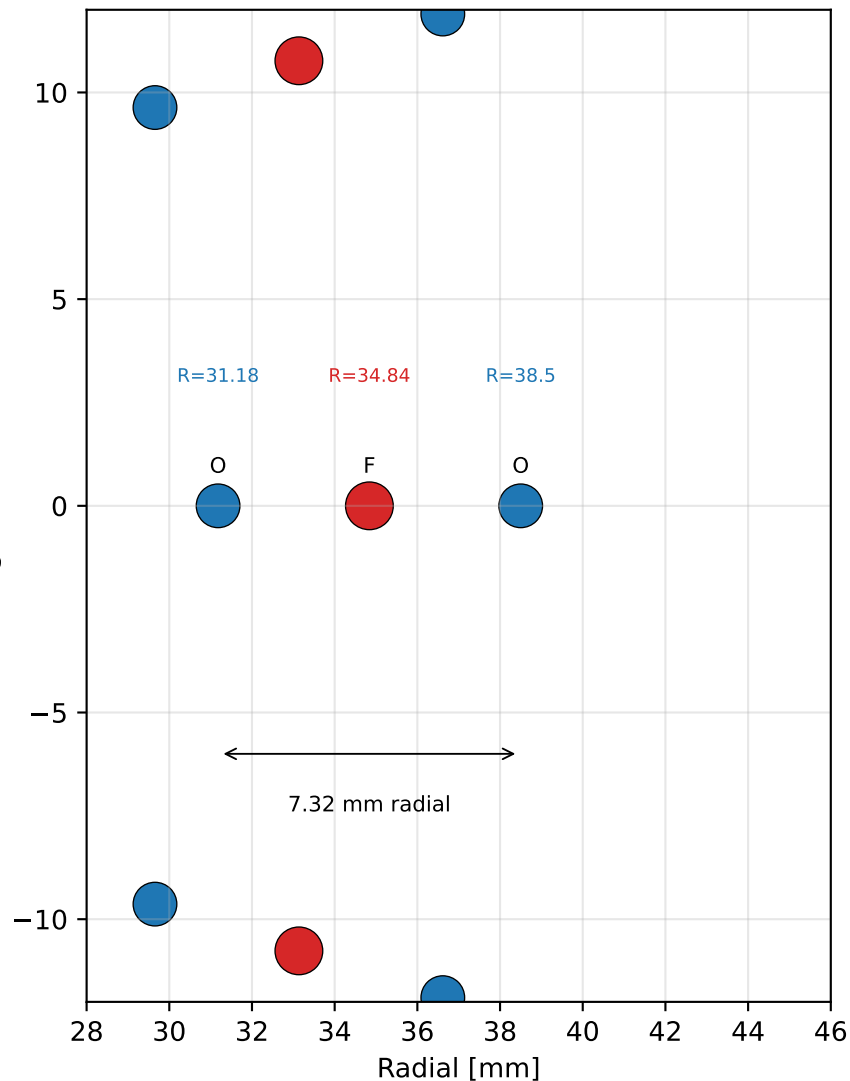
LOX : inlet (90 K) -> 20 face cooling channels
-> 2x LOX manifold rings -> 40 ox orifices
RP-1: chamber regen exit (~500 K) -> fuel ring
-> 20 fuel orifices (NO face-cooling duty)

60 orifices: 20 fuel @ $R=34.84$ + 40 ox @ $R=31.18/38.50$
(orifice glyphs 4x actual size for visibility)

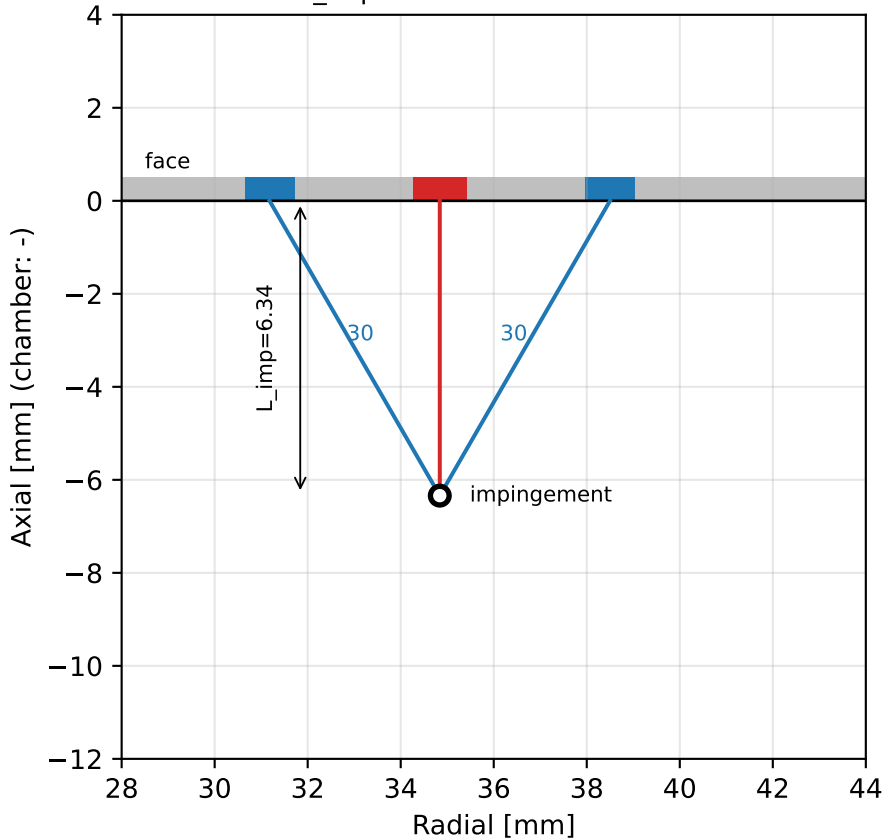


Page 3 — Single Element Detail (Element 1, $\theta = 0$)

Face view: 3 orifices per element on a radial line



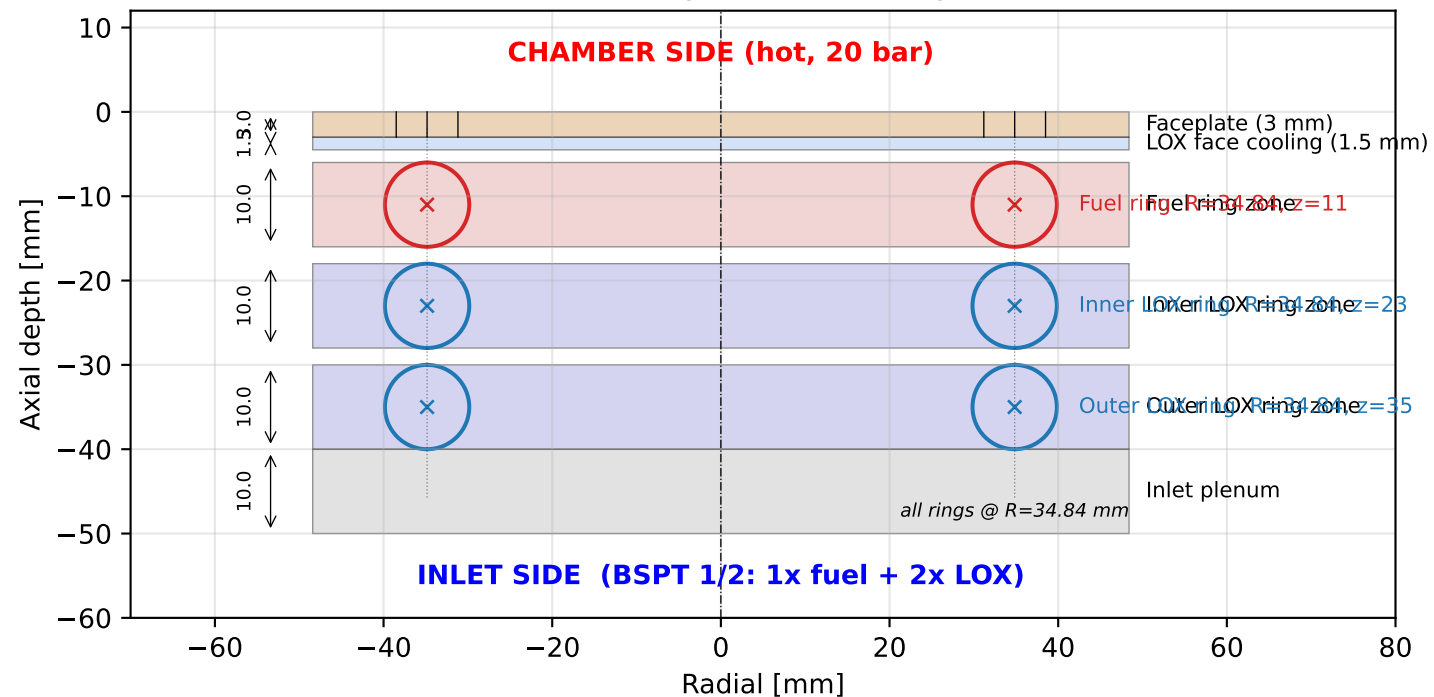
Cross-section: ox jets impinge on fuel jet
 $L_{\text{imp}} = 6.34$ mm below face



Page 4 — Injector Cross-Section (Axial Stack)

LOX flows: inlet -> face cooling -> LOX rings -> ox orifices

RP-1 flows: chamber regen exit -> fuel ring -> fuel orifices



Page 5 — Orifice Position Table (CAD Coordinates)

All 60 orifice centers (mm) — paste into CAD as sketch points

El	th[deg]	Inner ox x,y		Fuel x,y		Outer ox x,y	
1	0.0	+31.180,	+0.000	+34.840,	+0.000	+38.500,	+0.000
2	18.0	+29.654,	+9.635	+33.135,	+10.766	+36.616,	+11.897
3	36.0	+25.225,	+18.327	+28.186,	+20.478	+31.147,	+22.630
4	54.0	+18.327,	+25.225	+20.478,	+28.186	+22.630,	+31.147
5	72.0	+9.635,	+29.654	+10.766,	+33.135	+11.897,	+36.616
6	90.0	+0.000,	+31.180	+0.000,	+34.840	+0.000,	+38.500
7	108.0	-9.635,	+29.654	-10.766,	+33.135	-11.897,	+36.616
8	126.0	-18.327,	+25.225	-20.478,	+28.186	-22.630,	+31.147
9	144.0	-25.225,	+18.327	-28.186,	+20.478	-31.147,	+22.630
10	162.0	-29.654,	+9.635	-33.135,	+10.766	-36.616,	+11.897
11	180.0	-31.180,	+0.000	-34.840,	+0.000	-38.500,	+0.000
12	198.0	-29.654,	-9.635	-33.135,	-10.766	-36.616,	-11.897
13	216.0	-25.225,	-18.327	-28.186,	-20.478	-31.147,	-22.630
14	234.0	-18.327,	-25.225	-20.478,	-28.186	-22.630,	-31.147
15	252.0	-9.635,	-29.654	-10.766,	-33.135	-11.897,	-36.616
16	270.0	-0.000,	-31.180	-0.000,	-34.840	-0.000,	-38.500
17	288.0	+9.635,	-29.654	+10.766,	-33.135	+11.897,	-36.616
18	306.0	+18.327,	-25.225	+20.478,	-28.186	+22.630,	-31.147
19	324.0	+25.225,	-18.327	+28.186,	-20.478	+31.147,	-22.630
20	342.0	+29.654,	-9.635	+33.135,	-10.766	+36.616,	-11.897

CAD NOTES

- All 60 orifices on the face plane ($z = 0$)
- Fuel orifices: extrude axially, $\emptyset$ 1.154 mm, $L = 6.92$ mm
- Inner ox: extrude 30 deg toward centerline (+R tilt), $\emptyset$ 1.056
- Outer ox: extrude 30 deg toward centerline (-R tilt), $\emptyset$ 1.056
- Ox orifice length 6.34 mm, then stub channel to LOX ring
- LOX face channels: 20 radial slots 1.5×1.5 mm at angular positions $9 + n \cdot 18$ deg (between adjacent elements)
- LOX path: inlet -> face channels -> 2 LOX rings -> ox orifices
- RP-1 path: chamber regen exit -> fuel ring -> fuel orifices
(RP-1 does NOT cool the face -- already ~500 K, coking-limited)